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324631 (24)

BE (6th Semester)

Examination, Nov.-Dec., 2013

Branch : Electrical

FIBRE OPTICS

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each unit is compulsory. Attempt any two question from part (b), (c) and (d).

UNIT - I

Q. 1. (a) Define Numerical Aperture as applied to optical fiber. 2

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(b) Explain mode theory for circular wave guide.

(c) A multimode step index fiber is having following parameters :

V number = 76

Core diameter = 65 μm

Core refractive index = 1.5

Cladding refractive index = 1.4

Operating wave length = 1250 nm

Carrier frequency = 6×10^{14} Hz

Determine

(i) Numerical aperture

(ii) Solid acceptance angle

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(iii) Normalized propagation constant

(iv) Total number of mode.

(d) With the aid of suitable diagram explain vapour deposition technique used to fabrication of high quality silica fiber. 7

UNIT - II

Q. 2. (a) What is the function of photo detector. 2

(b) State the requirement of optical sources to be used with optical fibers. Explain the working principle of LED. 7

(c) Explain the working of PIN photodiode. 7

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(d) With aid of diagram explain working of Avalanche photo diode. 7

UNIT - III

Q. 3. (a) Define material dispersion. 2

(b) Explain different type of losses occurs in fibers. 7

(c) The core layer of optical fiber is made from pure silica material and cladding is made 10% less than that of core layer. The length of optical fiber is 10 km. Calculate 7

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(i) Fractional index difference

(ii) The delay difference between slowest and fastest modes.

(iii) The rms pulse broadening

(iv) The maximum bit rate

(d) Explain with the aid of diagram dispersion shifted fiber. 7

UNIT - IV

Q. 4. (a) What is the use of rise time budget in fiber optic system ? 2

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(b) With the aid of block diagram explain the working of coherent optical fiber communication system. 7

(c) Explain RZ and NRZ coding and their effect on bit rate. 7

(d) Write short notes on Eye Pattern. 7

UNIT - V

Q. 5. (a) What is WDM. 2

(b) Draw a block diagram of SONET and describe its working 7

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(c) Explain the working of EDFA ? Give comparison between semiconductor and ptical amplifier. 7

(d) With the help of suitable diagram explain ring topology and star architecture. 7

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